Comment on CMS-0042-NC: Health Technology Ecosystem RFI

June 3, 2025

Submitted electronically via regulations.gov

Re: Request for Information – Health Technology Ecosystem (CMS-0042-NC)

From: Dental Standards Institute (DSI)
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Dear Administrator Oz, National Coordinator Keane, and ASTP Director Posnack:

Dental Standards Institute (DSI) appreciates the opportunity to respond to the **Health Technology Ecosystem** Request for Information. DSI is a nationally recognized, ANSI-accredited standards development organization dedicated to advancing interoperability and patient-centered innovation in oral health care. We strongly support CMS and ONC's efforts to strengthen digital health infrastructure and would like to **underscore the critical need to include dental health information** in this ecosystem. In dentistry today, compliance with existing data-sharing laws (HIPAA portability and the 21st Century Cures Act's interoperability mandates) is lagging, hindering patients' rights and coordinated care. We urge **strong enforcement of these existing regulations in the dental sector** and **adoption of open standards** to close the medical-dental information gap.

Notably, DSI has developed the **ANSI/DSI EEDRA 1.1 standard**, commonly known as **THOR (Transferrable Holistic Oral Record)**, to facilitate dental data interoperability. THOR establishes **workflows allowing dental patients to access, store, and transfer their electronic dental records seamlessly, with full control over permissions**. We believe **recognition and promotion of the THOR standard** by CMS and ONC would be a pivotal step toward integrating oral health data into broader health IT. By enabling standardized, secure exchange of dental records, THOR **eliminates archaic technical barriers** and **paves the way for better collaboration among providers, improving patient outcomes**. In our comments below, we highlight how enforcing existing rules and leveraging standards like THOR can enhance patient access, data portability, and innovation in dental care. We have organized our feedback by the specific RFI questions (PC, PR, PA, TD, VB series) most relevant to our expertise, with numbered sub-responses corresponding to the RFI question parts where appropriate.

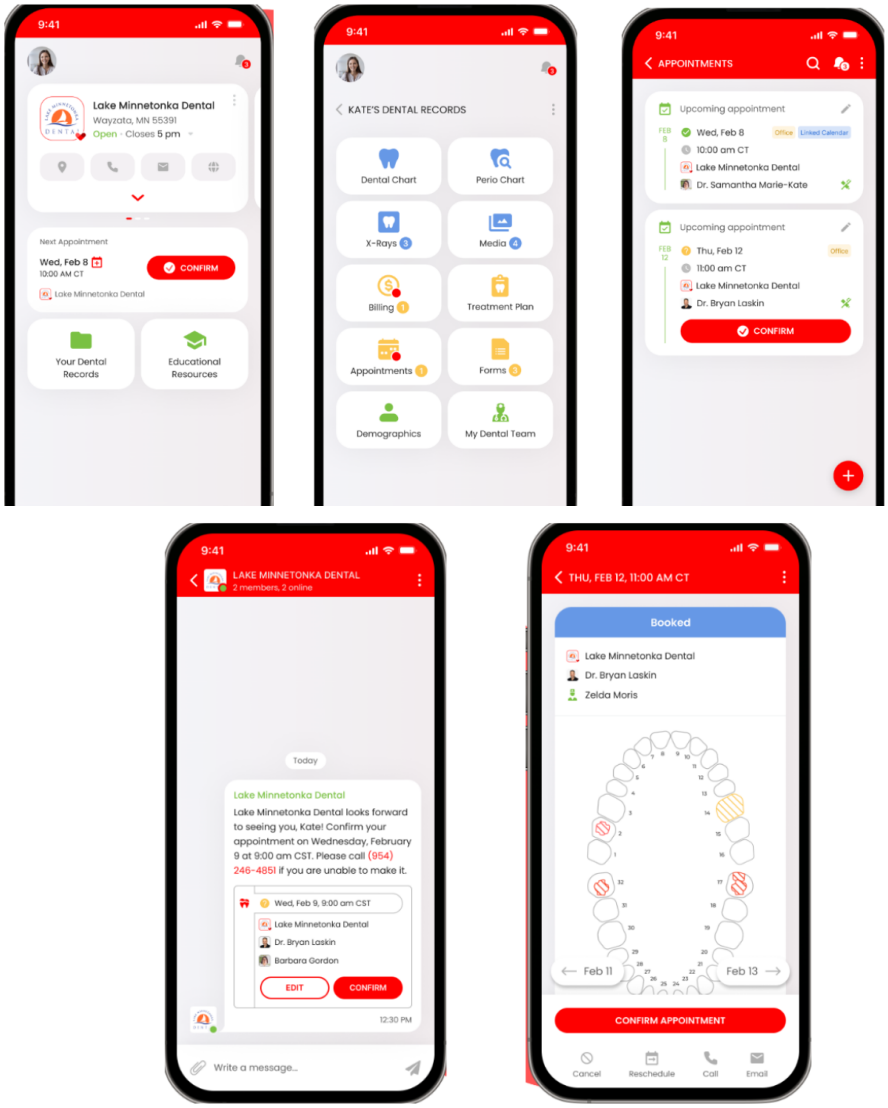
Patients & Caregivers (PC) Questions

PC-1. Care Apps To Manage Health Needs

We support digital health apps, like Cair, that help patients manage their dental records alongside key tools to manage their care. Most care navigation tools ignore oral health, leaving a major gap in patient understanding and actionability. Cair bridges this divide.

Top things patients should be able to do with digital dental health tools:

1. Instantly access and share dental records (x-rays, treatments, diagnoses).
2. View dental appointments.
3. Manage dental referrals.
4. Receive timely reminders for cleanings, follow-ups, and care coordination.
5. Compare treatment options, and costs.
6. Securely message their dental provider,
7. Understand how oral health affects systemic conditions.

PC-2. 

Easy

access to complete personal health information in one place:

Most patients, and especially in dentistry, do *not* have easy access to all their health information in a single portal or system. As a result, patients cannot view their dental x-rays, treatment plans, and history. This fragmentation undermines care coordination and patient empowerment.

> **(a) Benefits of unified access:** When patients can easily retrieve *all* their health data in, they experience improved continuity and quality of care. For example, a caregiver using a single portal to track an elderly parent's medications, and dental visits to catch potential issues (like a needed dental clearance before surgery) without delay. In practice, however, dental data is rarely included in such unified views and the data is almost always unavailable outside of the dental practice. **Where data access and integration has been achieved, the benefits include fewer repeated x-rays or procedures, faster information sharing during referrals, better treatment outcomes, and greater patient confidence.**
>
> **(b) Valuable use cases if unified access existed: Care coordination workflows** (between primary care physicians and dentists, or between general dentists and oral surgeons) would be far more efficient with a shared information access point. Patients repeatedly indicate that they would value such unified access; in fact, *100% of dentists* in one survey said they want to view patients' medical records, and *75% of physicians* want to see dental information; a testament that both patients and providers find cross-domain data valuable.
>
> **(c) Obstacles to accessing complete health info:** Several barriers prevent patients from having one-stop electronic access today. **Technical incompatibility** is a primary issue: dental practice management systems and medical EHRs use different formats and rarely communicate, so data isn't aggregated. **Data types** like dental radiographs and periodontal charts are often unavailable outside the dental office's software; patients usually get those via print or CD if at all. **Lack of interoperability standards adoption** in dentistry has meant even basic data (e.g., problem lists, allergies) typically do not transfer electronically. Additionally, **cultural and economic factors** play a role: some dental providers have been reluctant to share records, viewing patient data as a proprietary asset to retain business. This attitude, essentially a form of information blocking, leads to cumbersome processes for patients (requiring written requests, fees, or office visits to obtain records). Indeed, many patients have faced *"insurmountable challenges"* trying to get their dental records; the process to simply view or transfer records is often *"needlessly complex"* and **essentially impossible** for the average patient. Such obstacles persist despite HIPAA's guaranteed right of access.
>
> **We urge CMS to work with ONC on enforcing portability requirements in *all* care settings (including dental), and to promote standards (like DSI's THOR) that can**

bridge the medical-dental silos.

PC-5. Encouraging patient and caregiver interest in digital health products:

Digital health tools, like Cair, hold great promise for Medicare beneficiaries and caregivers – from mobile health management apps to personal health record aggregators – but interest and uptake remain limited in dentistry and other domains. **CMS and its partners can take several steps to spark greater patient engagement with these products:**

- **Education and Outreach:** CMS should collaborate with provider organizations, like DSI, and patient advocacy groups to **educate patients** on the availability and benefits of digital dental data aggregation apps, like Cair. Many dental patients are simply unaware that they could manage appointments, access records, or receive oral health coaching via an app. Outreach campaigns (via Medicare handbooks, 1-800-MEDICARE, or dental offices) highlighting success stories of digital health use can build interest. For example, a brief from CMS could show how using a medication reminder and dental care app together helped a patient improve both glucose control *and* oral health – emphasizing real-world benefits.

- **Data Availability and Integration:** Patients will be more interested in dental apps if those apps have **useful data and functionality**. CMS should ensure that the necessary health data is readily accessible to authorized apps (by enforcing existing regulations, interoperability and open APIs). This includes dental data that could greatly enhance a patient's understanding of their overall health, but today most apps lack dental information due to poor integration. By *"liberating patient data"* across the board (claims, dental visit information, etc.), CMS makes digital tools more compelling. In essence, **the content drives the interest**: when an app can show a patient all their intraoral scans *and* a reminder that they are due for a teeth cleaning, patients see value. Thus, a key strategy is rigorous implementation of interoperability rules so apps can pull comprehensive records via APIs.

- **Incentivize Provider Recommendations:** Often, patients trust their healthcare providers' advice on using new tools. CMS could encourage providers (including dentists) to recommend vetted digital health products. This might involve **including digital health usage in care coordination efforts or quality measures**. For instance, an accountable care program could measure how many patients use a patient portal or health app, creating an incentive for providers to promote these technologies. If dentists were required (or encouraged through CE credits, etc.) to inform patients how to access their electronic records or use a health app to track oral hygiene, it would boost adoption. Providers can frame these tools as extensions of care (e.g., "Use this app to monitor your healing after surgery and stay in touch with our office"), increasing patient uptake.

- **Trust and Quality Assurance:** CMS could play a role in **reviewing or certifying digital dental products** for efficacy and privacy, to increase patient trust. We do *not* advocate an onerous approval regime that stifles innovation; rather, a voluntary "CMS-Endorsed" label or a curated app library for Medicare beneficiaries could reassure users that certain apps meet quality, security, and accessibility standards. For example, CMS might evaluate personal health record apps on criteria like usability for seniors and proper handling of Medicare data. A light-touch **certification/endorsement** program would help caregivers and patients choose reliable tools (similar to how ONC's Certification Program gives confidence in EHR systems' capabilities). CMS could also coordinate with the FDA where apps verge into clinical decision support. In summary, CMS's role should be to **facilitate trust** (by setting basic standards and showcasing successful tools) rather than directly regulate every app's content.

In all these efforts, the messaging should be simple and empowering: dental data access can make patients' lives easier and outcomes better. CMS and partners might, for instance, highlight that **better data-sharing leads to better care**, e.g., "By using a secure health app, you and your doctors (and dentists) can share information instantly, avoiding delays in care". When patients and caregivers see clear personal benefit, their interest in using these products will naturally grow. **CMS should also ensure equitable access**, providing support for patients with low technical literacy or disabilities so that digital health benefits *all* populations.

(Regarding CMS's role in product review: we suggest a supportive, guidance role; establishing guidelines for efficacy and privacy, rather than a mandatory approval process that could slow down innovation. Aligning with existing frameworks (like FDA's digital health guidance and ONC's trust frameworks) is suggested.)

PC-7 Collecting Real-world Data

CMS should take immediate and targeted action to ensure that patients and caregivers can access, manage, and share their complete dental health information, just as they do with medical records. Dentistry continues to lag far behind in implementing basic patient access rights under HIPAA and the 21st Century Cures Act, despite clear regulatory requirements.

Here's what CMS can do now:

1. Enforce existing regulations in dentistry.
Almost all dental providers and EDR vendors still do not comply with HIPAA's right of access or Cures Act information blocking rules. CMS should increase oversight and issue clear guidance that these laws fully apply to dental records—including structured data, imaging, and clinical notes. Without enforcement, dental patients remain digitally disconnected.

2. Recognize tools that simplify access.
CMS should support platforms, like *Cair*, that give patients centralized, secure access to their dental records and allow them to share information with caregivers, providers, and care

coordinators. These tools reduce friction, empower families, and increase care continuity across settings.

3. Expand interoperability frameworks to include dental systems.

Incentivize or require dental EHRs to participate in TEFCA, use certified APIs, and support data export standards like THOR (Transferrable Holistic Oral Record). Without interoperability, access remains restricted.

4. Include dental access in quality and payment programs.
Add dental record-sharing capabilities as a performance measure in APMs, MIPS, and other quality programs. Patients should not have to beg for their records or be charged to access them.

Patients and caregivers need full access to dental data to make informed decisions, coordinate care, and manage oral-systemic health. CMS has the authority, and the responsibility, to ensure that dental data is as accessible, portable, and actionable as medical data.

PC-9. Expanding CMS data available via Blue Button 2.0 API:

The Blue Button 2.0 API has been a great step toward patient data access, but as noted, it currently includes only basic demographics, Medicare Parts A, B, and D claims. **Developers and patients would benefit from additional CMS data sources being made available via API**, to provide a more complete picture of beneficiary health. We recommend:

(a) Difficulties in accessing/utilizing these data today

Currently, developers face fragmentation. For Part C, there is no federal API. One must approach individual MA plans (which may not offer any API). Medicaid data is siloed by state and often not developer-accessible. Even within Blue Button's scope, some challenges include data latency (claims data may lag weeks to months behind real-time care) and complexity of claims coding (which can be hard for patients or apps to interpret without additional context). Technical obstacles include navigating multiple authentication systems for different payers and dealing with varying data formats. Policy obstacles include privacy concerns and restrictions on data reuse (some data may be sensitive, like substance use treatment claims, requiring special handling).

(b) Suggestions to improve the Blue Button 2.0 experience:

We suggest CMS continue to enhance the **developer-friendliness** of Blue Button. This could involve: more frequent data refreshes or real-time feeds for certain data (if feasible), improving documentation and sample code for integrators, and possibly providing a sandbox with synthetic data for easier testing. CMS might also implement a unified consent management so beneficiaries can authorize multiple data sources at once. For example, a patient could approve an app to access both their Medicare and

Medicaid data in one step, rather than separate processes, streamlining user experience. Additionally, expanding the data dictionary to give clearer descriptions (in lay terms) of claims entries would help app developers present information in patient-friendly ways.

(c) Non-CMS data that should be included:

We see a strong case for **integrating oral health data** into patients' digital health records. While not traditionally in CMS systems (except for limited cases like some Part C plans or specific Medicare-covered dental procedures), oral health information is critical to overall health management. For instance, knowing a diabetic patient's periodontal treatment history or a list of dental medications can be very relevant to their medical care. CMS could explore partnerships or standards to allow **patient-mediated inclusion of dental EHR data** into the ecosystem. One approach could be supporting standards like THOR, where patients obtain their dental records electronically and enabling those records to be uploaded or linked alongside Medicare data. While CMS doesn't generate that data, it can play a convening role to encourage common APIs and data exchange frameworks that bring it all together for the patient's benefit.

In summary, we urge CMS to expand Blue Button 2.0 to be a more **comprehensive health record API**, not just a Medicare claims API. This includes adding *all parts of Medicare* (A, B, C, D) and facilitating integration with other sources like Medicaid and potentially patient-contributed data. By filling these gaps, digital products for patients and caregivers can become far more useful and insightful.

PC-10. TEFCA's impact on patient access to health information:

TEFCA is in early stages of implementation, so **real-world impacts on patient access are limited but anticipated**. We address each sub-part:

(a) How TEFCA is helping advance patient access (examples):

Thus far, TEFCA's primary contribution has been to establish a unified trust framework across *Health Information Networks (HINs)*. As Qualified Health Information Networks (QHINs) come online, we expect patients will benefit from simpler connectivity. For example, consider a Medicare beneficiary who has records at various providers across the country (hospital in one state, specialist in another). Without TEFCA, the patient might have to access multiple portals or use apps tied to each provider's network. *With* TEFCA, a patient-facing app connected to one QHIN could theoretically retrieve that patient's records from any other QHIN participant network, using the common framework. In practice, one emerging example is that some regional HIEs (Health Information Exchanges) and national networks have **begun onboarding as QHINs** – a patient connected to one of these might start seeing more comprehensive data. We acknowledge that concrete patient-facing use cases under TEFCA are still ramping up.

However, one early real-world scenario might be **individual access services** offered by certain QHINs, where a patient can query "the network of networks" for their records. For instance, if a patient uses a personal health record app that is an authorized participant in TEFCA, they could discover and retrieve information across multiple states/providers in one action, something not previously feasible. This is the sort of cross-network access TEFCA is poised to enable.

(b) Suggested changes to TEFCA to improve patient access: We recommend **strengthening TEFCA's requirements and capabilities for consumer access**.

First, TEFCA should ensure that **Individual Access Services (IAS)** are a *must-have* offering by QHINs (not just optional). Every QHIN should be capable of responding to patient queries for their own data, and this should be operationalized early. CMS and ONC could incentivize QHINs to prioritize this use case. Second, **expand the data scope within TEFCA**: currently it's geared toward clinical summary documents; to truly help patients, TEFCA should facilitate exchange of a wider variety of data (images, full text notes, and importantly *dental records* or other traditionally siloed data). We suggest ONC consider incorporating emerging standards (like imaging APIs, dental data standards such as THOR) into TEFCA's umbrella as optional but encouraged capabilities. Another suggestion is to create a **user-friendly patient interface** at the national level – for example, a CMS/ONC-sponsored "Patient Access Gateway" that helps patients connect with QHIN services in plain language. Finally, ongoing governance: patients should have representation in TEFCA's governance to voice needed changes and ensure the framework remains patient-centric.

(c) High-impact use cases if implemented through TEFCA:

We believe one especially valuable use case is **medical-dental integration**: for example, if TEFCA facilitated a use case where a dentist could query for a patient's medical conditions and medications through a network (with patient permission), it could significantly improve safe dental treatment for medically complex patients. Conversely, a primary care provider under TEFCA could query if their diabetic patient has been receiving periodontal care. These cross-domain use cases have *significant impact on outcomes* and are currently under-served by existing exchange arrangements.

(d) Standards currently working well to advance access/exchange:

The health IT community has coalesced around some key standards that are proving effective. **HL7® FHIR® (Fast Healthcare Interoperability Resources)** is foremost among these – it underpins modern APIs like Blue Button and many patient-facing apps. FHIR's widespread adoption in certified EHRs has enabled patients to use apps (e.g., Apple Health) to access and aggregate their data from different providers using a

consistent RESTful API. That is a major leap forward for patient access. Additionally, **Blue Button 2.0's use of FHIR** for claims data is an example of a standard API working well to give beneficiaries insight into their utilization. We also see **DICOM** imaging standards as crucial for sharing radiographs and other images. When dental or medical images adhere to DICOM, it's much easier to exchange them. In summary, FHIR-based APIs combined with standardized data formats (USCDI for core data, DICOM for images, etc.) are working well when implemented, and should continue to be the backbone of interoperability efforts.

(e) Standards not in wide use that could improve access/integration:

We want to highlight the **ANSI/DSI THOR standard** (DSI EEDRA 1.1) here. THOR is a newly published standard specifically designed to enable **patient-centric exchange of dental health records**. It is not yet widely implemented (given its recent release), but it has enormous potential to improve data integration between dentistry and the rest of healthcare. By providing standardized workflows for patients to *directly access and transfer their dental records*, THOR can bring the dental field into parity with medical interoperability. We urge CMS and ONC to consider recognizing or endorsing THOR as a valid standard for EHR vendors and HIEs to adopt, which would dramatically increase its uptake. In addition to THOR, there are other promising but underused standards: for example, **HL7 FHIR Bulk Data (Flat FHIR)** for efficient population-level data sharing – not yet widely used, but it could transform how organizations exchange data for care coordination and analytics. **Patient-centered data elements** like those being proposed in USCDI+ (e.g., detailed oral health data, social needs data) are also currently not widely standard, but including them would improve integration of those domains. Another area is **standardized dental diagnostic codes** (currently, dentistry primarily uses CDT procedure codes; incorporation of SNOMED/SNODENT CT dental concepts could improve semantic interoperability). While those coding standards exist, they're not yet widely required or used.Greater adoption could enhance data exchange quality. Lastly, **personal health device integration standards** (like IEEE 11073 for remote monitoring devices) could be more broadly used to incorporate patient-generated health data into records. In sum, there are strong standards waiting in the wings. By shining a light on them and including them in certification or guidance, CMS/ONC can markedly improve data access. THOR, in particular, deserves attention as it addresses a gaping hole in interoperability today (dental records).

(f) Redundant standards/protocols that should be consolidated:

Yes, we do observe redundancies that could be streamlined. A prime example is the coexistence of older document-based exchange standards alongside newer API-based ones – for instance, many networks support both IHE XDS/XCA query-based document exchanges and also FHIR-based APIs. Maintaining multiple pathways increases complexity. Over time, **consolidating on modern, API-driven approaches (like FHIR)**

would reduce burden on developers and providers. We suggest that CMS and ONC encourage the phase-out of proprietary or legacy protocols when a robust standards-based alternative exists. For instance, if a vendor has a custom data export format but also the capability to use a standardized FHIR API, CMS could incentivize using the FHIR and deprecating the custom method. Another redundant area is identity management – there are multiple parallel identity proofing systems (some networks use custom patient identity matching versus others adopting common services). Consolidating around **common trust frameworks for identity (e.g., NIST IAL2 standards)** would help patients not have to manage different credentials for each exchange network. Also, within TEFCA, if overlapping schemas or exchange methods are allowed, it might be prudent to pick one preferred approach for each exchange purpose to simplify implementation. Generally, we favor **unification**: pick the best, most future-proof standard and encourage everyone to use it, rather than maintaining duplicative channels. This reduces confusion and costs. For example, encouraging all health data queries to eventually use FHIR RESTful calls (with standardized resources) could replace the mélange of SOAP, REST, and batch file exchanges that currently exist. We acknowledge some legacy will persist for backward compatibility, but setting a clear direction (and timeline) for convergence on one method is important.

(g) Alternatives outside of TEFCA for widespread patient access:

While alternatives exist and provide some widespread access (through apps and networks), none yet achieve the *universality* and consistency that TEFCA aspires to. We believe TEFCA is needed to tie these threads together and fill gaps (especially when patients get care from multiple regions or different IT systems). Outside TEFCA patients rely on a patchwork: multiple portals, multiple apps, or HIE request processes. It's better than nothing and improving due to Cures Act requirements, but it's still fragmented. TEFCA can greatly streamline this by establishing one overarching highway for data.

PC-11. Role of Health Information Exchanges (HIEs) in advancing patient access:

(a) Value, availability, and accuracy of HIE-shared data:

Almost all HIEs still have limited participation from certain provider types; notably, **dental providers rarely contribute to or retrieve data from HIEs** today, meaning HIE datasets often omit oral health information altogether. A statistic illustrating current limitations: fewer than half of dental providers co-located in health centers could input information into the shared EHR system, and 88% of medical/dental providers in those settings couldn't update each other's treatment plans. This indicates that even in integrated settings, shared data is incomplete – a challenge HIEs should work to overcome (by bringing more providers, including dental, onto the network).

(b) Suggested changes to HIEs for better patient access:

Broaden HIE participation. HIEs should strive to onboard *all* relevant healthcare entities in their community, which means reaching out to smaller practices, dental clinics, behavioral health, long-term care, etc. CMS can assist by incentivizing or requiring participation through policy (for example, tying Medicaid or Medicare participation to at least reporting data to an HIE, where available). Including dental records in HIE data is a change that would notably enhance patient access to a truly complete health record.

(c) High-performing HIE models to propagate:

One model example is the concept of a **community health record** provided by some HIEs. For instance, HIEs in states like Indiana and Maryland have developed portals where a patient or authorized caregiver can log in and see a consolidated health record drawn from all participating providers. These have been praised for improving care transitions. An HIE like CRISP (Chesapeake Regional Information System for our Patients) in Maryland/DC, for example, offers patient access features and integrates data from hospitals, physicians, and some long-term care facilities. Propagating models where HIEs serve patients *directly* (not just providers) would advance nationwide patient access. The **common traits** of high-performing models are: comprehensive data sources, easy patient access (web or app), and strong stakeholder trust in data accuracy.

(d) Ongoing role of HIEs vs vendor networks and broader frameworks:

We view HIEs as complementary to initiatives like vendor networks and TEFCA, rather than obsolete. HIEs often have deep regional knowledge and relationships, allowing them to curate data and fill gaps that nationwide vendor networks might miss. For example, a local HIE might include small community clinics or public health data (immunizations, disease registries) that a vendor-specific network doesn't capture. Amidst private vendor exchange networks and TEFCA, HIEs can function as **hub translators and data quality stewards**. They can take data from various sources (including non-standard ones) and normalize it for use, for example as Toothapps normalizes dental EDR information from within proprietary software platforms in dentistry. HIEs also often support **analytics, alerts, and care coordination services** (like notifying a primary care doctor when their patient is admitted to any hospital in the region). Those value-added services mean HIEs are more than just data pipelines; they actively facilitate improved care. So, we foresee HIEs continuing to play an integral role: acting as **bridges and translators** between disparate systems, championing community-level interoperability, and focusing on comprehensive data that includes traditionally siloed sectors (like dental). The presence of vendor networks and TEFCA

will push HIEs to evolve (some consolidation may occur), but their mission of complete, accurate, and accessible records for all stakeholders (including patients) remains vital.

PC-13. CMS Encouraging Information Blocking Complaints to ASTP/ONC Information Blocking Portal

CMS can and should encourage patients and caregivers to report information blocking; especially in dentistry, where the problem is widespread and under-reported.

Most dental patients are unaware they even *have* a legal right to access their complete records, or that it's a violation when they're denied access, charged unreasonable fees, or told they can't have x-rays or clinical notes. This behavior by dental providers and vendors routinely violates HIPAA and the 21st Century Cures Act.

To increase reporting and drive change, CMS should:

1. **Launch targeted education campaigns for dental patients** that explain their rights under HIPAA and the Cures Act, and how to report noncompliance.

2. **Embed clear "report a violation" links** in patient portals, CMS dental benefit communications, and care navigation apps like *Cair*.

3. **Require dental EHRs and providers to notify patients** of their access rights and the ONC complaint portal during care transitions or record requests.

4. **Collaborate with organizations like DSI and ADA** to distribute accessible complaint templates and patient advocacy materials.

Impact:
Greater reporting would *accelerate compliance* and transparency in dental data exchange. **Today's low complaint volume gives the false impression that patients aren't being harmed. In reality, many dental vendors openly block access, confident there will be no consequences.**

By making the complaint process visible and easy, CMS would empower patients, expose bad actors, and create the pressure needed to enforce interoperability across the dental industry, without harming legitimate data exchange.

Reporting is a catalyst for progress. Encouraging patients to speak up, especially through modern platforms that already aggregate and manage dental data, is essential to finally bringing dentistry into alignment with national health data standards.

Provider (PR) Questions

PR-3. Importance of making *all* EHR data (regardless of format) interoperable:

We assert that it is *very important* that **all data within an EHR – structured or unstructured, scanned or text – be accessible for exchange** in both medical and dental. Incomplete data exchange can lead to serious gaps in care quality. Particularly in dentistry and other specialties, critical pieces of information (like imaging, specialist consult letters, or detailed procedure notes) may reside as PDFs or scanned documents. If these are not exchangeable, a referring provider or care team might miss key facts. **Every data format (from a discrete allergy entry to a scanned tooth chart or a panoramic x-ray image) should be available to share** when needed for patient care. This is true for big city hospitals and solo rural clinics alike; in fact, rural and underserved areas often rely on external referrals and thus need even more robust sharing to compensate for limited local resources.

To address sub-parts:

(a) Current challenges in accessing different data formats:

A major challenge is that **unstructured data (scans, free-text, images)** often doesn't flow through standard interfaces. For example, a rural dentist might only have paper or scanned records of a patient's prior treatments; if that patient moves, the new dentist often just gets a fax or nothing at all. **Proprietary formats** exacerbate the issue: some dental x-rays are stored in vendor-specific formats that other systems can't read. Another challenge is the sheer logistics, such as large image files (like DICOM radiographs or CT scans) being bandwidth-intensive, and small clinics may struggle to upload/download them quickly, especially in rural areas with limited internet.

(b) Impact on patient care quality:

When parts of the record aren't accessible, the quality of care suffers. Providers end up making decisions with incomplete information. For example, if a physician refers a patient to a dentist but the **scanned anticoagulation therapy note** doesn't transfer, the dentist might be unaware of a bleeding risk, potentially endangering the patient. Conversely, if a dentist's findings (say, signs of oral cancer noted in a narrative) aren't shared back to the medical record because they were in a free-text note, a primary care provider might miss an early warning. **Care continuity and safety depend on all data traveling with the patient.** Also, inaccessible data often leads to redundant tests (if an image isn't available, an X-ray might be retaken, meaning unnecessary radiation and cost) or missed diagnoses.. Enabling access to *all* EHR data will improve diagnostic accuracy, reduce treatment delays, and avoid adverse events stemming from information gaps.

(c) Technical barriers to full data accessibility:

One barrier is that **EHR systems sometimes segregate data**; not all are exposed to the exchange interface. Additionally, **computational resources** and performance matter.

(d) Cost or privacy implications of making all data formats interoperable:

There are certainly considerations here. On cost: enabling interoperability for all formats might require upgrades to software and hardware. Providers or vendors might incur costs to implement and maintain these capabilities (e.g., secure image exchange servers, additional cloud storage for attachments, etc.). However, these costs must be weighed against the savings from improved efficiency (less redundant imaging, fewer manual record requests) and improved outcomes. From a policy perspective, **information blocking rules already discourage charging unreasonable fees for data exchange** but there could be legitimate cost recovery concerns if, say, an office is asked to digitize years of paper charts for exchange. **CMS could help by subsidizing interoperability infrastructure or mandating cost-free exchange as a norm** (to prevent "fees" exceptions from being abused).

On privacy: with more data flowing, particularly detailed notes and behavioral or dental records, **privacy safeguards are paramount**. Ensuring that **only authorized parties access the data** and that patients have control is critical. Another aspect is **patient consent**: particularly for records like psychotherapy notes (excluded from sharing) or certain reproductive health info, etc., which patients might want extra control over. Systems will need to accommodate these nuances (possibly tagging data with sensitivity flags and honoring consent choices). But generally, **we believe privacy can be protected *while* sharing comprehensive data**, through robust access controls, audit logs, and patient education.

PR-7. Strategies for CMS to support providers in sharing high-quality, timely, comprehensive data (while mitigating burden):

From a dental industry perspective, we see several strategies CMS can implement to help providers (medical and dental) make more data available interoperably, without adding undue burden:

1. **Enforce and Expand Existing Interoperability Requirements:** The groundwork from HIPAA and the 21st Century Cures Act is already in place. Patients have the right to access their records, and providers are prohibited from blocking information. However, enforcement in dentistry has been minimal. **CMS can work with OCR and ONC to actively enforce these rules** for all healthcare providers, including dentists. This means

ensuring providers are aware that, for example, they *must* provide records electronically upon request and cannot refuse legitimate data exchange. Clear consequences (penalties or public reporting) for non-compliance will motivate providers to prioritize data sharing. Many providers currently ignore these mandates due to lack of accountability; stronger enforcement flips the incentive to compliance.

2. **Incentivize Interoperability via Payment and Quality Programs:** CMS has powerful levers in its quality payment programs. We recommend embedding interoperability goals into existing provider incentive structures. For dental providers, who largely don't participate in MIPS, CMS might coordinate with other payers (MA plans, Medicaid, etc.) to create incentives. The core concept is to make **data sharing financially rewarding** (or at least, making non-sharing gradually financially punitive). This flips the current situation where some practices feel they have a financial disincentive to share (fear of patient attrition).

3. **Leverage Existing Workflows to Automate Data Sharing:** A key way to avoid "extra burden" on providers is to make interoperability seamless; part of the routine, not an additional task. CMS can push EHR vendors and standards bodies to develop workflows where data sharing happens in the background. For example, fulfilling a patient record request via an API should be as easy as one click approval by the provider, not a lengthy process. CMS could highlight and promote **use cases where existing clinical workflows double as data sharing events**. Some are already defined in CMS programs; for instance, the "closing the referral loop" measure (which requires sending a summary of care) is essentially an interoperability workflow. The idea is to align what providers already want to do (coordinate care) with the data sharing mechanism, so it doesn't feel like a separate bureaucratic step.

4. **Reduce Documentation Burden by Harmonizing Reporting Requirements:** Often, providers feel burdened when new data reporting demands are layered on. CMS should strive to **reuse data for multiple purposes** to avoid double entry. If a provider documents structured data in the EHR for clinical care, that same data should satisfy quality reporting, public health, and interoperability needs. For example, if a dentist enters caries risk and periodontal status in their system for clinical reasons, that could automatically feed quality measures (like a metric for managing diabetic patients' gum health) and also be available to share with other providers or patients. Additionally, CMS could align any new interoperability metrics with existing provider goals. Providers are already motivated to improve outcomes and patient satisfaction; therefore highlighting how interoperability directly contributes to those goals will harness their intrinsic motivation. In essence, **make interoperability a solution to current pains**, not an extra chore. One tangible example: providers hate redundant paperwork, so CMS could establish interoperable data exchange, removing requirements for filling duplicate forms, thus *reducing* burden in practice.

In summary, **CMS should use both the "carrot and stick"**; enforce the rules that require sharing (stick), and offer incentives and automation to make sharing the path of least resistance (carrot). Done right, providers will see interoperability not as an added burden but as a means to improve care and achieve recognition. We specifically stress that dentistry needs to be included in these strategies: many dental providers have not yet embraced electronic data sharing, so targeted outreach (via dental schools, associations, large group practices) combined with enforcement of information blocking prohibitions will be needed to bring them on board. This will greatly increase the volume and quality of data available for patient care across the ecosystem.

PR-12. Reviewing information blocking exceptions to further data access and competition:
The Information Blocking Rule (45 CFR Part 171) currently defines several exceptions (for privacy, security, preventing harm, infeasibility, etc.) under which an actor's refusal to share data *might* be permitted. We believe ASTP/ONC should **carefully reconsider whether all these exceptions, as written, are necessary or if they inadvertently provide loopholes that hinder legitimate access, exchange, and use of electronic health information (EHI)**. Scaling back overly broad exceptions could indeed further the goals of seamless access and a more competitive market for health IT and services.

Two exceptions in particular stand out for potential revision: the **Fees exception and the Licensing exception**. As it stands, these allow actors to charge fees for data exchange or to require licensing of interoperability elements, within certain conditions. In practice, some EHR developers and networks have interpreted these in a permissive way; charging high "API fees" or making onerous licensing demands for integration. This can price smaller providers or startup software developers out of the market, effectively **stifling competition** and perpetuating data silos. We recommend ONC tighten these exceptions – for example, by clarifying that only **reasonable, nominal fees** are allowed (truly cost-based, not profit-generating), and perhaps capping fees for certain common types of data exchange (like patient-directed API access should likely be free of charge to the patient and minimal cost to developers). Similarly, for licensing, ONC could specify that any required license for interoperability elements (such as code sets or API use) must be non-exclusive and fair/reasonable. By reducing the financial friction, more players can access data and innovate, which ultimately benefits patients.

Another area is the **"Content and Manner" exception**. While flexibility is good, we worry that this can be abused to *unnecessarily* limit data sharing. For example, a provider might claim they can't share via a standard API and instead offer to print out records or burn a CD (citing the content and manner exception). This defeats the goal of near-frictionless exchange. ONC should consider revising this exception to *mandate* the use of **widely available standards** when they exist. If a patient asks for their records via a FHIR API, and the provider has any certified technology, they should not be allowed to refuse that and insist on a slower method unless it's truly impossible. Essentially, narrow the "infeasibility" wiggle room here – if an actor

has the capability to meet the requested mode (or a reasonably close alternative), they should do so. The bar for claiming infeasibility or alternative manner should be high and well-justified.

Furthermore, ONC might evaluate whether all eight exceptions remain appropriate as we move beyond USCDI-limited info to full EHI sharing. For example, the **"Preventing Harm" exception** is important (to avoid truly harmful disclosures), but could it be clarified to prevent misuse (like a blanket assumption that patients will be upset by lab results and thus withholding them – which would be an unreasonable interpretation)? Each exception's conditions should be scrutinized for ambiguity. If any exception is frequently cited in complaints as a reason data was blocked, that's a candidate for refinement or removal. The goal should be that exceptions are *rarely invoked* and only in genuine cases, not as routine justifications.

In the dental context, we've seen that some vendors or providers lean on perceived legal exceptions (like misinterpreting privacy laws) to justify not sharing records. Tightening the federal exceptions and providing clear guidance (e.g., "HIPAA *requires* you to share with patients even if they owe you money") removes excuses. It also fosters **market competition** by leveling the playing field – new entrants (like an independent personal health app or a small dental software vendor) can access data under uniform rules rather than being blocked by incumbent players citing broad exceptions.

In conclusion, we support revising the information blocking exceptions to be **narrower and more specific**, ensuring they cannot be exploited to delay or deny legitimate data access. The focus should remain on the default: **information *should* be shared "without special effort" whenever permissible by privacy law**, and exceptions are truly exceptional. This will encourage a more vibrant ecosystem where data flows to wherever it can improve care or spur innovation, without artificial roadblocks.

PR-12. ASTP/ONC Revising Information Blocking Exceptions

Yes, ASTP/ONC should revise certain information blocking exceptions to close loopholes that are routinely exploited in dentistry, where compliance with the Cures Act remains dangerously low.

Key recommendations:

1. **Tighten the "Infeasibility" exception.**
   Many dental vendors claim technical or cost-based infeasibility to deny access or delay record transfers. This is often used as a default excuse to avoid implementing APIs or export features. The exception should require *objective justification*, not vague assertions, and carry time-bound thresholds.

2. **Clarify and narrow the "Content and Manner" exception.**
   Dental providers and vendors often refuse to provide structured or usable formats, citing a lack of common standards. In reality, open formats like THOR now exist. This exception should require providers to *default to available standards* unless clearly

documented and justified.

3. **Re-express the "Fees" exception to prohibit abusive pricing models.**
   Dental EHRs frequently charge excessive fees to extract or share data, essentially holding patient information hostage. CMS/ONC should more clearly define what constitutes "reasonable cost" and prohibit fees that restrict access or stifle third-party competition.

4. **Limit the overuse of the "Security" exception.**
   Security concerns are sometimes weaponized to block authorized patient access or third-party integration. This exception should be rephrased to protect against abuse, with an emphasis on scalable safeguards, not denial of access.

Impact:
Revising these exceptions will prevent bad-faith actors, especially in the dental sector, from hiding behind policy ambiguities. It will level the playing field for innovative tools like *Cair*, improve patient access, and promote market competition by removing artificial barriers to entry created by legacy vendors.

ONC and ASTP must modernize the information blocking exceptions to reflect the real-world misuse happening in underserved sectors like dentistry. Clearer language, tighter enforcement, and fewer loopholes will drive the access, innovation, and equity the Cures Act intended.

PR-14. Encouraging providers to submit information blocking complaints (and potential impacts):

Here are actions CMS can take to encourage complaint submissions:

- **Awareness Campaign:** CMS should widely publicize *how and where* to report information blocking. This could include email bulletins to providers, segments in CMS webinars, and coordination with professional societies to spread the word. For dentists, CMS might work with DSI, the ADA, specialty groups, and state dental boards to distribute guidance on the info blocking rule and the availability of the complaint portal. Simply making more providers conscious that "If you can't get the data you need, or someone is making it unreasonably hard, you can (and should) report it" will drive up reporting. Right now, lack of awareness is a huge barrier.

- **Simplify the Reporting Process:** Ensure that the online portal is user-friendly. CMS could create a short, simple form for common scenarios (like "I requested records from X and they took 3 months/charged exorbitantly/didn't send structured data"). If providers find the reporting burdensome, they won't follow through. Perhaps CMS can also allow *batch or third-party facilitated reports*; e.g., a dental association could submit a summary

of multiple members' complaints in one go.

Impact of encouraging more complaints: In the short term, increased complaint submissions might surface the extent of information blocking. CMS and ONC could see a spike in cases to review. This is actually beneficial, as it shines light on what has been a mostly hidden problem. Yes, it could mean more administrative work to process complaints, but it provides the raw data needed to target enforcement and guidance. As enforcement ramps up, we anticipate a **positive deterrent effect**: actors (whether EHR vendors, hospitals, or others) will be much less likely to engage in blocking behavior if they know providers are empowered and willing to report them. Ultimately, this should **advance data exchange**: the threat of being called out and penalized will lead organizations to prioritize compliance (e.g., responding to record requests faster, reducing unnecessary fees, etc.).

Could there be any negative impacts? Possibly, some might worry that an avalanche of complaints could create an adversarial atmosphere or that providers might misuse the portal (reporting competitors frivolously). We believe these are minor risks; ONC can filter and investigate complaints appropriately. The overall effect of transparency and accountability should be strongly positive. Data exchange will improve as the **culture shifts from tolerance of blocking to expectation of openness**. Providers submitting complaints are effectively advocates for their patients' continuity of care.

One specific note: Many dentists have historically been outside the federal health IT spotlight, so they may not even realize that refusing to send records or delaying transfers is unlawful. By encouraging reporting, CMS also indirectly educates the whole community that such blocking behavior is not acceptable. The mere act of promoting the complaint mechanism serves notice that regulators are serious about interoperability in *all* sectors, dentistry included.

Payer (PA) Questions

PA-2. Encouraging payers to accelerate API implementation and utilization:

CMS has already taken significant steps by requiring Medicare Advantage, Medicaid, and zTo further accelerate and broaden the use of APIs by payers (for patients, providers, and payer-to-payer exchange), CMS can consider several approaches:

- **Strict Enforcement of Existing Requirements:** First and foremost, ensure that all CMS-regulated payers (MA organizations, state Medicaid agencies, CHIP, QHP issuers) are in full compliance with current API mandates. This means conducting oversight or audits to verify that these payers not only have the APIs up and running, but that they meet **availability and performance standards**. If any payers are lagging (for instance, some state Medicaid have waivers or delays), CMS should push them to catch up. Clear

enforcement signals (including possible corrective actions or penalties for non-compliance) will maintain momentum.

- **Expand Scope to More Payers and Data Domains:** Within its own programs, CMS could extend API requirements to **dental benefit plans** that are associated with MA or Medicaid.
- **Promote API Utilization through Incentives and Publicity:** Simply having an API doesn't mean it's being used. CMS can help by **publicizing which payers have functional, easy-to-use APIs** and perhaps creating a "scorecard" or recognition program.

CMS has laid down rules; now it must ensure follow-through and cultivate an environment where **using the APIs becomes standard practice**. Through enforcement, expanded requirements (including dental coverage areas), and incentives , CMS can speed up both implementation and active use of these APIs. The result will be patients more easily accessing their data, providers obtaining information seamlessly from payers, and payers themselves operating in a more transparent, competitive marketplace where members expect data portability.

Technology Vendors, Data Providers, and Networks (TD) Questions

TD-2. Stimulating developer interest with CMS data (additional data, obstacles, other APIs):

(a) Valuable additional data through CMS APIs:

More types of data from CMS would enable richer features. One high-value addition would be **Medicare *Part C* (Medicare Advantage) data**, including encounter data, capitated plan data, and supplemental benefits usage (e.g., dental, vision, transportation services if covered by MA). Also, **quality and performance data** from CMS programs could spark innovation. For example, an API that provides a beneficiary's providers' quality metrics or a hospital's star rating might be used in decision support tools for patients choosing providers.

(b) Data sources valuable alongside Blue Button 2.0:

For patient-facing apps, it would be valuable if Blue Button could integrate (or at least reference) data from **health information exchanges or CommonTrust networks** where clinical data might reside.

(c) Obstacles to accessing these sources today:

Inconsistent standards are an issue: Blue Button uses FHIR for claims, but a state might use a different format, making it harder to combine. Lastly, small innovators may face **cost barriers**: while CMS APIs are free, connecting to other networks might involve fees or legal agreements that startups find daunting.

(d) Other APIs CMS/ONC should consider for unleashing innovation:

We propose a **"Dental Records API":** As stressed, dental data is a missing piece. CMS could collaborate with ONC and DSI to encourage an API standard for dental electronic records, analogous to how USCDI covers core medical data. Whether via FHIR profiles or leveraging the THOR standard, enabling apps to pull a patient's dental history (e.g., recent procedures, x-rays, periodontal measurements) would be groundbreaking. CMS might not host this data, but it could support a **policy that dentists adopt APIs**.

TD-4. Encouraging use of open, standards-based APIs over proprietary APIs:

Open, standards-based APIs are critical for scalability and innovation, whereas proprietary APIs create vendor lock-in and barriers.

In the dental domain, historically vendors typically use proprietary data export formats which trap patient information. Encouraging those vendors to adopt standards such as FHIR or DSI's THOR workflows is vital. CMS could facilitate a pilot where a few dental systems implement a standard API and demonstrate how easily a patient can transfer records. Using that success to challenge others: "If these vendors can do it, all can."

TD-10. Further steps to implement the Cures Act's API Condition of Certification (all data, no special effort):

We suggest:

- **Mandate Full EHI Scope in APIs (Beyond USCDI):** As of now, many certified APIs only guarantee **USCDI data elements**. The law calls for "all data elements of a patient's EHR" to be accessible. ONC should set a clear timeline or standard (perhaps USCDI v3+ or a special "EHI FHIR profile") by which certified APIs must encompass *full* EHI. For example, by 2026, certification could require FHIR support for clinical notes, diagnostic images, and other data classes beyond the core.

- **Address "Special Effort" Loopholes:** The condition is clear that data should be accessible without special effort. ONC might further clarify what "special effort" means in practice; e.g., patients or third-party apps shouldn't have to pay fees, navigate through proprietary gateways, or endure long delays. Essentially, a certified API should be plug-

and-play for a consumer app *by design*.

- **Expand Certification Program to More Developer Types:** Right now, the API condition binds “health IT developers of certified health IT”. That leaves out many systems (like most dental EHRs, some specialty EHRs, etc.) that chose not to certify under the ONC program. As a concrete step, ONC could hold a “certification expansion” workshop for dental software, inviting DSI and vendors to map how they could comply with Cures Act APIs. This would bring more products into compliance with the spirit of the law.

- **Active Surveillance and Enforcement:** Implement a mechanism to continually verify that certified APIs are functioning as required. CMS/ONC could support development of a **testing tool or monitoring service** that regularly pings these APIs (with dummy data) to ensure they are returning the full data and not experiencing outages or restrictions. Any non-conformance (like a vendor silently not supporting certain data types) should trigger corrective action (up to certification suspension). Providers could be engaged to report if they discover missing data via APIs. That feedback loop can inform ONC to tighten requirements.

TD-12. CMS endorsement of non-CMS data sources

Yes, CMS should endorse non-CMS data networks, especially in sectors like dentistry, where CMS infrastructure is limited or nonexistent. Independent networks are essential to ensure patients have access to their full health history.

Recommended Endorsement Criteria:

1. Patient Access & Control – Enables easy viewing, sharing, and permission management.

2. Use of Open Standards – Supports recognized frameworks like THOR for structured dental data.

3. Security Compliance – Meets HIPAA and data protection requirements.

4. Data Completeness – Offers comprehensive records, not partial summaries.

5. Vendor Neutrality – Allows open participation from EHRs, apps, and innovators (e.g., *Cair*).

Endorsing standards-based, patient-first networks outside of CMS would accelerate dental data interoperability, support innovation, and close major gaps in equitable health data access.

TD-13. Opportunities and advancements with APIs accessing the entirety of a patient's EHI (and obstacles/tradeoffs):

Providing API access to the **entirety of a patient's electronic health information (EHI)** (not just a core subset) opens up a world of possibilities for improving healthcare and spurring innovation. We outline some key opportunities, primary obstacles, and the trade-offs between focusing on USCDI vs. full EHI:

- **New Opportunities and Advancements:** When APIs can liberate all of a patient's data, we move closer to true **patient-centered, holistic care**. For example, a comprehensive personal health application could aggregate everything – from clinical diagnoses and lab results to dental records and social history – and use AI to provide personalized health insights. A concrete advancement might be **integrated chronic disease management:** imagine an app for diabetic patients that not only tracks glucose readings but also pulls in periodontal exam data from the dentist (since gum disease can affect blood sugar). This could generate tailored advice like, "Your blood sugar tends to improve after dental cleanings; consider scheduling one if you're overdue." Researchers and clinicians could discover such correlations only when full datasets are available via API. Another advancement is **streamlined transitions of care**: with full EHI access, when patients switch providers or levels of care, they can bring *all* their data. No more relying on summaries that omit details, the new provider's system (or the patient's care coordination app) could pull the complete prior record (medication fill histories, imaging files, specialist notes). This leads to safer and quicker onboarding of new patients and less practice burden. **Innovations in research and public health** are also ripe: if patients donate their full data to studies via API (a model of patient-mediated research), datasets will be richer, enabling discoveries (for instance, linking oral health markers with heart disease outcomes) that were not possible when data was siloed. Additionally, **market innovation** thrives: full data liquidity lowers entry barriers for new care models (like telehealth services can have full context when seeing a patient, or community health workers can use apps that show everything needed to intervene appropriately).

- **Primary Obstacles to full EHI API access:** The obstacles are non-trivial. **Data volume and complexity** is a big one. Entire EHRs can be gigabytes of data per patient (especially with imaging and long histories). Transmitting and handling that via API requires robust infrastructure and could be slow or prone to timeouts without careful design (bulk data protocols, etc.). **Heterogeneity of data formats**: not every piece of EHI is in neat FHIR resource form. There will be custom fields, older formats, or narrative text which APIs need to deliver in a usable way. Ensuring that "all EHI" doesn't turn into "a dump of incoherent data" is a challenge; standardization must catch up to encompass more data types. **Privacy and consent** issues become more acute. Certain parts of the record may require extra patient consent or have legal protections (behavioral health, reproductive health, etc.). APIs will need fine-grained consent management so patients can say "yes, share everything except perhaps mental health notes or that one clinic's records" if they so choose. On the provider side, **cultural resistance** could be an obstacle: some clinicians worry about giving patients unfettered

access to full records (concerns about misinterpretation of raw notes, or litigation fears). These attitudes are changing but still present; education and possibly policy may be needed to overcome any deliberate withholding. **Technical obstacles** include ensuring patient identity matching across systems for comprehensive pulling of data. If records aren't linked correctly, you might get incomplete or wrong data.

- **Trade-offs between USCDI (core dataset) and full EHI:** USCDI offers a standardized, relatively compact set of data elements that everyone agrees are important for exchange (demographics, problems, meds, etc.). Focusing on USCDI first (as the industry has done) made sense to get interoperability off the ground. The trade-off is that **USCDI is easier to implement and ensures a baseline of data everywhere**, but it leaves out a lot that could be clinically important. For instance, USCDI might tell me a patient's medication list and diagnoses, but not the full context of *why* a certain therapy was chosen or the narrative of how the patient is doing – which might be in the doctor's notes. Full EHI access means nothing is left out, which is better for completeness but **harder to process**. Many apps or clinicians may feel overwhelmed by the volume and complexity of full EHI; a firehose versus a garden hose of data. There's also the reality that some data in "full EHI" may not be immediately useful: for example, device logs, audit trails, intermediate workflow data. Including them could clutter the meaningful info. So, a trade-off is **signal vs noise**: USCDI is mostly signal; full EHI will include some noise that apps have to sift through. Another trade-off is in **performance and cost**: restricting to USCDI means smaller payloads, quicker responses, and less storage needed on the receiving end. Full EHI might strain systems (for example, pulling decades of a patient's history every time might be inefficient if an app only needs the last year – developers will have to be smart about that). **One approach is tiered or hybrid:** use USCDI as the common core everyone handles by default (for interoperability ease), but have the option to get more (full EHI) when needed. The Cures Act essentially pushes toward that anyway: baseline easy access to core data, with the ability to go broader/deeper on demand. We view USCDI as the foundation, the minimum, not the ceiling. Trade-offs can be managed with thoughtful API design (like filtering parameters, so an app can request "just USCDI" or "full dataset" per its needs) and leveraging use case specific standards, such as THOR, for currently less available standardized data (so the "full" data is still structured where possible).

We encourage CMS/ONC to keep pushing the envelope, ensuring that eventually, **when a patient wants their entire record, they get it via API easily**, and when an app or provider needs the nuance beyond the common data, it's there to be used for the patient's benefit.

TD-18. Information blocking experiences from a technology vendor perspective, and further discouraging provider information blocking (including actor categories lacking disincentives):

(a) Examples of practices constituting information blocking (vendor perspective):

As an SDO interfacing with many technology stakeholders, we have observed numerous practices that likely constitute information blocking:

- **Lack of API or Data Export Features:** Some health IT vendors (including in dental software) simply do not provide any reasonable means to extract patient data electronically. For instance, a dental EHR vendor might not have a patient export function or API, effectively trapping the data unless the office goes through manual printing or pays for a custom extract. This non-responsiveness to interoperability needs is a form of blocking by omission, especially if the vendor has been asked by clients for a data interface and has delayed or refused. We've heard of cases (and experienced times) where providers request their vendor enable some data sharing (say, with a referral management system) and the vendor just stalls indefinitely, which is **non-responsiveness** in practice. And in many cases, they flat out refuse, expecting that there will not be any punitive damages placed on them (as detailed above).

- **Excessive Fees for Data Access:** Another example is vendors who technically offer an integration or data export, but only at a very high cost. One dental practice reported that when they decided to switch to a different software, their current vendor quoted an exorbitant fee (thousands of dollars) to release a copy of the patient records database. This **price gouging** for data handover is effectively information blocking. It uses the Fees exception in a possibly inappropriate way to dissuade the practice from switching (or from giving the data to another vendor).

- **Proprietary Formats and Unusable Responses:** Even when data is provided, some vendors do it in a way that hinders use. For example, a practice might request an export of all patient records; the vendor might provide it but in a proprietary, non-standard format or a gigantic unstructured PDF that is nearly impossible to import elsewhere. Such an **unusable response** technically fulfills "giving the data" but not in a practical sense

- **Interoperability by Exclusive Partnership:** Another observed practice, common with dental EHRs, is when a tech vendor will only integrate with certain favored partners and not with others, even when there's customer demand. For example, an EHR might have an API, but only allow access if you're using that vendor's preferred app or network, but if a new third-party app tries to connect, the vendor throws up obstacles (lengthy approval processes, additional fees, technical roadblocks). This creates a walled garden that blocks innovative entrants. We have seen vendors who build "app marketplaces" but keep tight control, effectively **blocking unapproved apps** from accessing patient data on their platform.

These examples cover both vendor behaviors and provider behaviors. They show that despite the legal requirements, there are myriad ways some actors still impede the free flow of information.

(b) Additional policies to discourage providers from information blocking:

While the ONC rule defines and prohibits information blocking, one gap has been effective **disincentives for certain providers**. CMS could implement or advocate for several policies:

- **Monetary Penalties or Reimbursement Tie-ins:** Currently, unlike health IT developers and some networks (who face certification or civil monetary penalties), many providers (except those in certain CMS programs) lack clear penalties. CMS could work with HHS to institute a schedule of **monetary fines** for willful information blocking by providers, as authorized by Cures Act for those under its purview. This could be scaled by severity or repeat offenses. Additionally, CMS could make compliance with information blocking a condition of participation in Medicare/Medicaid. For example, **attestation of no information blocking could be required for provider enrollment or revalidation**, similar to how providers attest to not being excluded or to having compliance programs. Non-attestation or false attestation could result in actions on their participation status.

- **Public Reporting/Transparency:** CMS could coordinate a public reporting mechanism where substantiated cases of information blocking by providers are published (name and shame). If a dentist or health system is found to consistently refuse sharing data, making that known publicly (e.g., on Hospital Compare or Physician Compare sites) would pressure them via market/public opinion to improve. Patients and referring providers would avoid those who block, creating reputational consequences.

- **Strengthen Education and Support for Providers:** Some providers block out of misunderstanding or fear (e.g., mis-believing HIPAA prevents sharing, or not knowing how to share safely). CMS, in partnership with ONC, should intensify education: clear guidance that *patient access and appropriate provider exchange is not just allowed, but expected*. Possibly create an "Information Sharing Handbook" for providers with scenarios: e.g., when a patient requests data, you must provide it within X days electronically – here's how.

- **Patient Empowerment:** Empowering patients to be intermediaries can help. For example, CMS could promote that patients use their rights; e.g., encourage beneficiaries to use Blue Button or Cair app access and to report to CMS if a provider refuses to cooperate with an app connection. If providers know patients are vigilant and CMS is backing patients up, they'll be less likely to refuse.

(c) Actors lacking information blocking disincentives:

- **Healthcare providers:** Many providers, especially those **not in federal incentive programs (e.g., dentists)** effectively have *no specific disincentive* yet beyond the general legal prohibition. They're unlikely to face any consequence unless OCR/ONC acts on a complaint, which has been rare so far. We highlight **dentistry** as a category: dentists are clearly defined as healthcare providers in the law, but since they weren't eligible for EHR incentives and aren't typically subject to CMS quality payment, they

have no CMS-driven penalty for blocking. We strongly recommend implementing disincentives for these provider types. For example, perhaps HHS could work with Congress or use existing authority to extend some form of penalty, even if nominal, to any provider found to block (like a fine or public notice, as discussed). CMS could also use its influence via Medicaid (as many dentists participate there), requiring states to enforce info blocking compliance in provider agreements.

Value-Based Care Organizations (VB) Questions

VB-2. Key Technologies to be better integrated into APM requirements.

Recommendations for CMS Integration into APMs:

1. **Require inclusion of dental data in population health and risk models.**
   Use open formats like THOR to integrate dental records into medical APM infrastructure, enabling smarter analytics and preventive care strategies.

2. **Support AI and care coordination tools that include oral health.**
   Endorse platforms (e.g., *Cair*) that unify patient-facing and provider-facing tools with embedded dental insights.

3. **Tie APM incentives to data completeness and usability.**
   Encourage EHRs and providers to expose structured dental data for quality measurement, care navigation, and patient engagement.

4. **Use oral health status as a quality metric.**
   Include measures such as untreated caries, periodontal disease tracking, or preventive dental visit adherence in risk-adjusted models.

Conclusion:
APMs can only succeed when powered by complete, connected data. Integrating dental information through standards-based tools will enhance AI, improve risk models, and make patient engagement more meaningful. Especially in value-based ecosystems.

VB-3. Essential health IT capabilities for value-based care arrangements:

We agree with the examples given in the RFI (care planning, event notifications, data normalization, quality measurement, claims data access, attribution and patient matching, remote device integration, patient engagement tools) and would add/emphasize a few capabilities, especially drawing from a dental integration perspective:

- **Connected Medical-Dental Health Records:** One capability that has proven valuable but is often missing is **connectivity of dental data into overall care management systems**. For value-based arrangements targeting whole-person health, knowing a

patient's oral health status and dental treatment history is essential. For example, undiagnosed dental issues can drive up medical costs (ER visits for dental pain, complications in diabetic control, etc.). An essential IT capability is the ability to **incorporate dental EHR data alongside medical data**. This could manifest as shared access to dental records for care coordinators or a data feed from dental clinics into the ACO's analytics platform. Having this capability enables more comprehensive care planning and risk stratification.

- **Population Health Analytics with Predictive Modeling:** Another essential capability is robust analytics to identify rising-risk patients and evaluate interventions. For example, an ACO should have tools that can predict hospital admission risk not just from medical history but also incorporating dental issues (e.g., frequent dental abscesses could be a sign of uncontrolled diabetes, contributing to risk). So the IT should allow combining multi-source data to generate risk scores and actionable insights. Advanced analytics, often AI-driven, can highlight, for instance, that patients with both periodontal disease and diabetes in the registry need targeted outreach, as such insights improve outcomes and reduce costs.

- **Interoperable Remote Monitoring and Telehealth:** VBC thrives on proactive management. Capabilities like integrating **remote patient monitoring device data** (blood pressure cuffs, glucose monitors, even teledentistry tools that monitor oral health remotely) are increasingly important. IT systems should ingest these data streams and flag out-of-range values to the care team. The IT should seamlessly incorporate telehealth encounters (including virtual dental consults for preventive care or triage).

- **Care Coordination Workflow Integration:** A practical capability is having IT that supports the actual workflows of care coordination; e.g., referral management systems that ensure when a PCP refers to a dentist, the referral loop closes with report backs.

VB-11. Interoperability challenges encountered in implementing value-based care programs:

Value-based care programs, by their nature, rely on comprehensive data sharing and coordination. Several interoperability challenges frequently arise:

- **Incomplete Data Integration Across Providers:** In many value-based arrangements, participants include a range of providers (primary care, specialists, hospitals, dental and behavioral health providers). **Integrating data from all these sources is challenging**. In pilot programs integrating oral health, they found that dental clinics couldn't easily send data to the ACO's data warehouse due to lack of interface, so oral health events were under-reported. This fragmentation makes it hard to coordinate care and measure outcomes accurately.

- **Patient Identification and Matching:** Value-based orgs often have to pull data from multiple sources (claims, EHRs, HIEs). **Accurate patient matching** is a perennial

challenge. If records aren't correctly linked, you either fragment a patient into two in your database (missing a portion of their care) or mistakenly merge two different people. Both are dangerous for managing care and measuring performance. Variations in identifiers (Medicare ID vs. internal IDs, etc.) and data quality (typos in names/DOB) complicate this. ACOs have spent significant effort on improving matching algorithms and still report difficulty, especially when including post-acute or community data sources that don't share a common identifier. In dentistry, implementing the ODIN ID, as part of the ADA Standard 1111, ODIN, could help to address this issue within dentistry.

In closing, Dental Standards Institute appreciates CMS and ASTP/ONC's attention to these critical issues. Enforcement of data portability and anti-blocking provisions in *all* corners of healthcare, including dentistry, is overdue and essential. We strongly support initiatives to **integrate dental data into the broader health IT infrastructure**, recognizing that the mouth is part of the body and oral health information is vital for truly patient-centered care.

By promoting and adopting standards like **THOR for dental interoperability**, CMS can ensure that the dental community catches up with interoperability advancements, thus empowering patients to access and share their complete health records effortlessly. We also advocate for robust measures that hold all actors accountable for open data exchange, thereby unleashing innovation in digital health tools and improving care coordination.

DSI stands ready to assist in these efforts; whether in refining standards, educating stakeholders, or piloting new data exchange models in dentistry. We are optimistic that with the policies and ideas discussed, from reinforcing HIPAA/Cures Act compliance in dentistry to fostering seamless API ecosystems, we can create a health technology ecosystem that truly serves patients and providers.

Thank you for considering our comments. We commend CMS and ONC for their leadership and welcome any further collaboration to advance connection across the medical-dental continuum and beyond.

Sincerely,

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